Penetration Testing Report – OWASP Juice Shop

Tested by: Saloni Kumari
Date: 08 May 2026
Target: http://localhost:3000

==========

1. Executive Summary

A penetration test was conducted on OWASP Juice Shop, an intentionally vulnerable web application. Multiple vulnerabilities were identified and successfully exploited including SQL injection, Cross-Site Scripting (XSS), exposed sensitive directories, and authentication bypass. This report documents the findings, exploitation steps, and remediation recommendations.

==========

2. Scope

- In-scope: OWASP Juice Shop running on localhost port 3000
- Out-of-scope: Any other systems or real-world websites

==========

3. Methodology

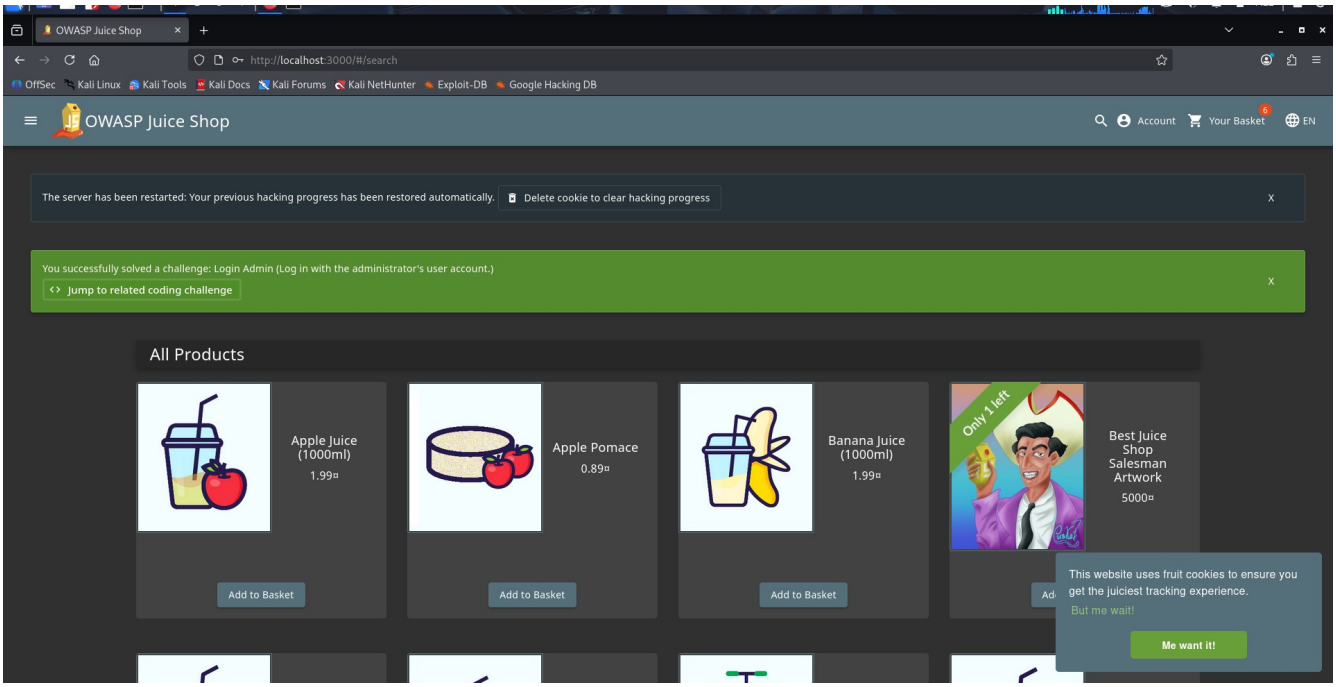
The following tools and techniques were used:
- SQL injection for login bypass
- Manual JavaScript payload injection for XSS
- Dirsearch for hidden directory discovery
- Burp Suite and curl for request manipulation (attempt)
- Browser testing for FTP directory exposure

==========

4. Findings & Exploitation

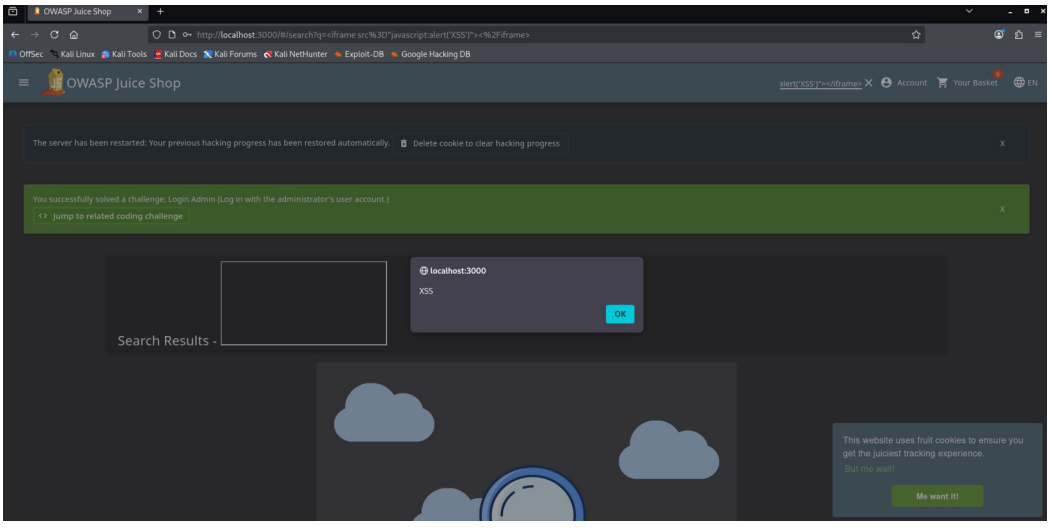
4.1 SQL Injection – Login Bypass

- Payload: admin' OR '1'='1' --
- Impact: Attacker can log in as admin without password



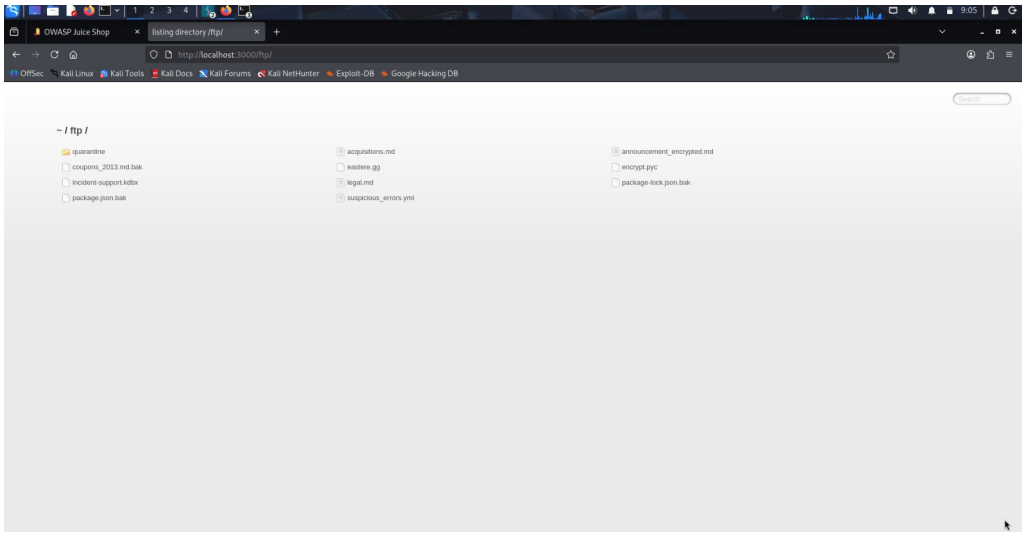
4.2 Cross-Site Scripting (Reflected XSS)

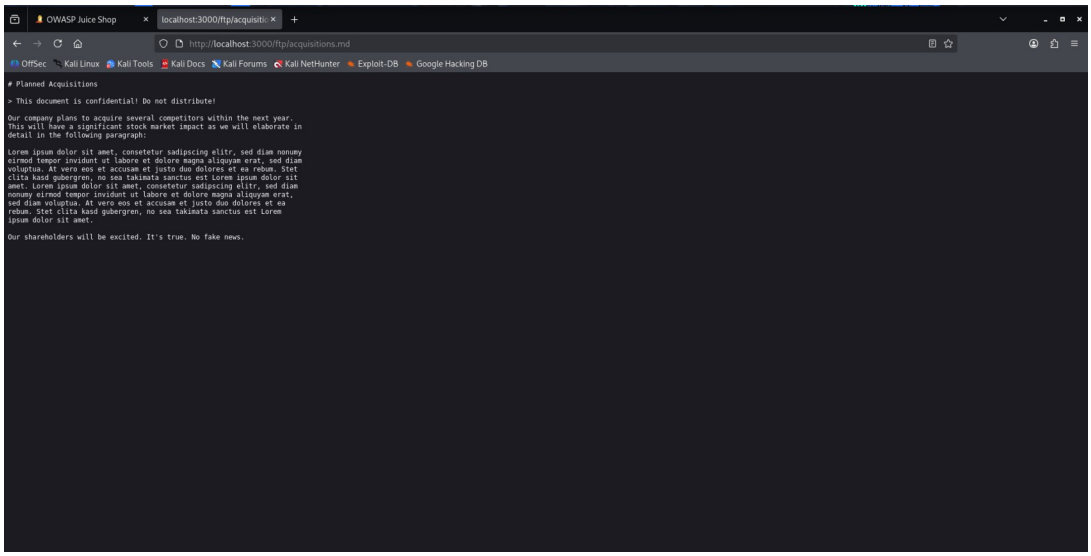
- Payload: <iframe src="javascript:alert('XSS')"></iframe>
- Impact: Execution of malicious script in victim's browser



4.3 Exposed /ftp/ Directory (Information Disclosure)

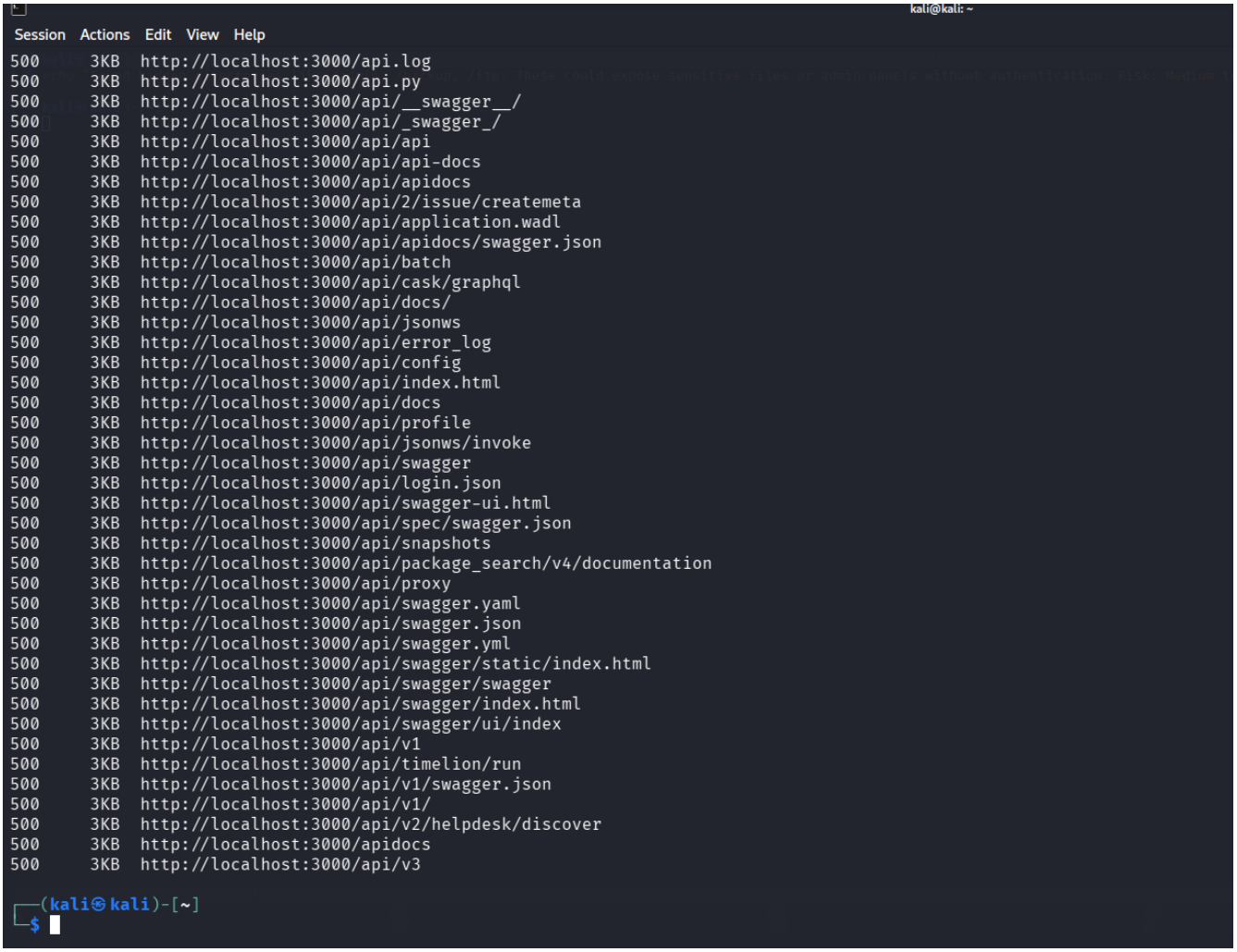
- Steps: Navigate to http://localhost:3000/ftp/
- Sensitive files found: acquisitions.md, package.json.bak, coupons_2013.md.bak
- Impact: Attacker can download sensitive internal files





4.4 Hidden Endpoints (Dirsearch)

- Discovered: /api/config, /api/error_log, /admin, /backup
- Risk: Potential exposure of configuration and admin panels



==========

5. Risk Assessment

Finding	Serverity	Likelihood
SQL Injection Login Bypass	Critical	High
Reflected XSS	Medium	Medium
Exposed/ftp/directory	Medium	High
Hidden admin/API endpoints	Low-Medium	Medium

==========

6. Remediation Recommendations

- SQL Injection: Use parameterized queries / prepared statements.
- XSS: Implement output encoding and Content Security Policy (CSP).
- Exposed /ftp/: Disable directory listing, add authentication, move sensitive files out of web root.
- Hidden endpoints: Restrict access via authentication or IP whitelisting.

==========

7. Conclusion

The application has several security weaknesses that could be exploited by an attacker. Immediate action is recommended for SQL injection and exposed directory issues. Regular security assessments should be conducted.

==========

Report prepared by: Saloni Kumari
Date: 08 May 2026